

BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Q2	Parameters	Q1	Q2
IS	8e-17	3.7e-17	MJC	0.53	0.18
BF	128	81	XCJC	1	0.5
NF	1	0.9944	CJS	0	0
VAF	17	27.7	VJS	0.75	0.75
IKF	0.18	0.98	MJS	0	0
ISE	3.3e-15	2.53e-10	FC	0.37	0.5
NE	1.48	30	TF	6e-12	5e-12
BR	9.05	12.3	XTF	11.9	0.2
NR	1.05	1.002	VTF	9.55	0.1
VAR	4.3	2.9	ITF	1.78	0.001
IKR	0.009	7.29	PTF	69.1	0
ISC	4e-15	1.35e-15	TR	1e-9	1e-11
NC	1.5	1.215	EG	1.11	1.11
RE	0.8	1	XTB	0	0
RB	11.1	8	XTI	3	3
RBM	2.46	4	KF	0	0
IRB	0.02	0.001	AF	1	1
RC	7.5	13.9			
CJE	0.415e-12	0.49e-12			
VJE	0.68	0.83			
MJE	0.53	0.18			
CJC	0.102e-12	0.12e-12			
VJC	0.29	0.6			

(1) Gummel-Poon Model

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

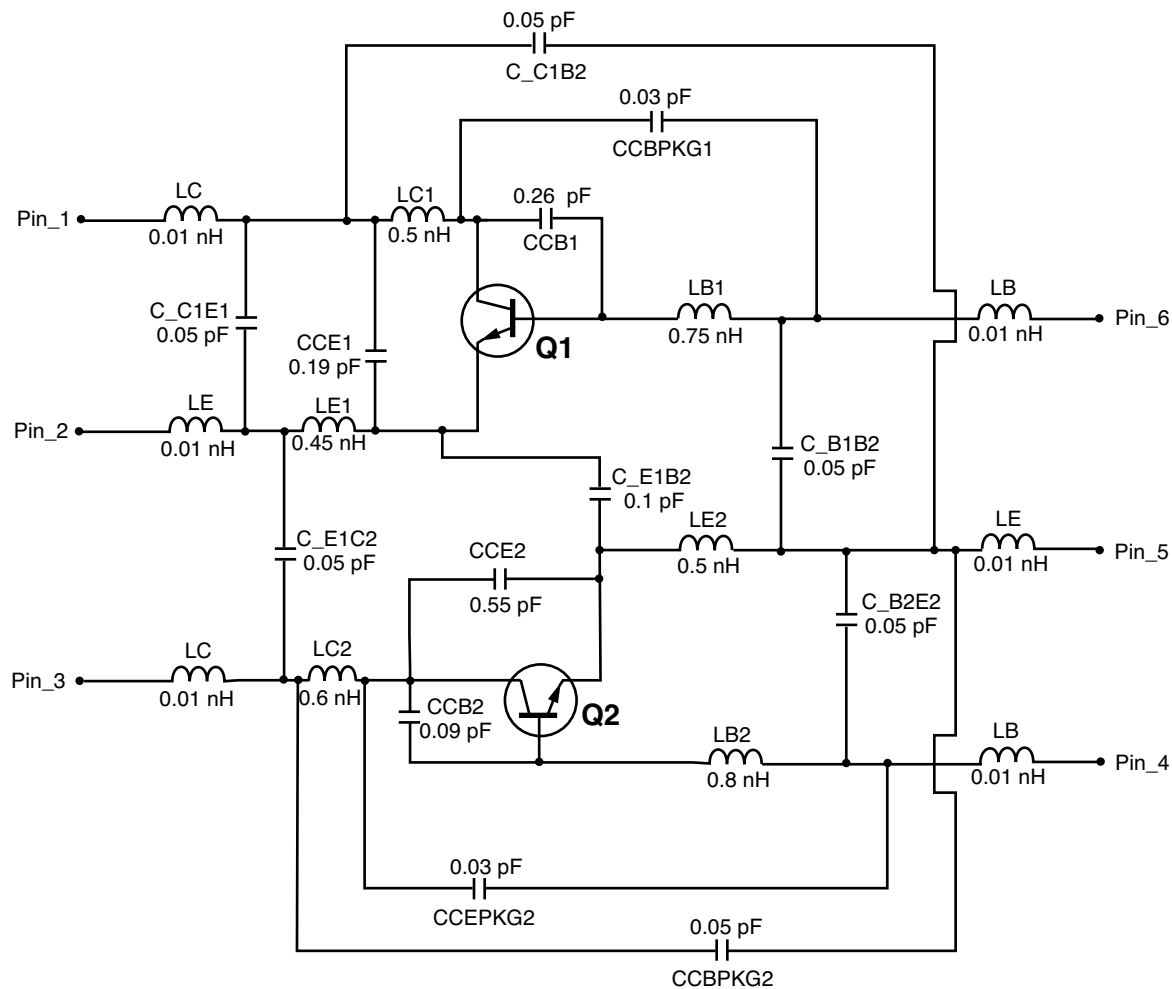
MODEL RANGE

Frequency: 0.1 to 4.0 GHz
 Bias: $V_{CE} = 0.5 \text{ V to } 2 \text{ V}$, $I_C = 0.5 \text{ mA to } 10 \text{ mA}$
 Date: 04/01

Note:

This nonlinear model utilized the latest data available. See our Design Parameter Library at www.cel.com for this data.

SCHEMATIC



MODEL RANGE

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